

Colab interface showing a Python script for a Journey Time Prediction System using Linear Regression.

Browser tabs: internship level 1(1,2).ipynb - Co, Ask Gemini

URL: colab.research.google.com/drive/1hW8UZh9mGZeKsu4nUWAv3bPAT21tArK#scrollTo=y3TBMalqIDDj

Commands: + Code, + Text, Run all, Saving...

RAM, Disk, Toggle header visibility

```
[31] ✓ Os
# LEVEL 6 - Journey Time Prediction System using Linear Regression
import pandas as pd
import matplotlib.pyplot as plt
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error, mean_squared_error
import math

df = pd.read_csv("Modeling_Dataset.csv")

X = df[["Total_Distance", "Number_of_Stops"]]
y = df["Journey_Duration"]

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

model = LinearRegression()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)

mae = mean_absolute_error(y_test, y_pred)
rmse = math.sqrt(mean_squared_error(y_test, y_pred))

print("===== MODEL PERFORMANCE =====\n")
print("Mean Absolute Error (MAE) :", round(mae, 2))
print("Root Mean Squared Error (RMSE) :", round(rmse, 2))
```

Variables, Terminal

✓ 12:12 AM Python 3

Colab interface showing a Jupyter Notebook with Python code for plotting Actual vs Predicted Journey Duration.

Browser tabs: internship level 1(1,2).ipynb - Colab

URL: colab.research.google.com/drive/1hW8UZ9mGZeKsu4nUWAv3bPAT21tArK#scrollTo=y3TBMalqIDDj

Code:

```
[31] print(prediction.head(10))

prediction.to_csv("Final_Predictions.csv", index=False)

plt.figure(figsize=(8,6))

plt.scatter(
    prediction["Actual_Journey_Duration"],
    prediction["Predicted_Journey_Duration"]
)

minimum = min(
    prediction["Actual_Journey_Duration"].min(),
    prediction["Predicted_Journey_Duration"].min()
)

maximum = max(
    prediction["Actual_Journey_Duration"].max(),
    prediction["Predicted_Journey_Duration"].max()
)

plt.plot(
    [minimum, maximum],
    [minimum, maximum],
    linewidth=2
)

plt.title("Actual vs Predicted Journey Duration")
plt.xlabel("Actual Journey Duration (Minutes)")
plt.ylabel("Predicted Journey Duration (Minutes)")
plt.grid(True)
```

RAM: 100% / 16GB, Disk: 100% / 100GB

Variables: {} Terminal

12:12 AM Python 3

Colab interface showing a Jupyter Notebook cell with Python code for plotting and predicting journey duration.

Browser tabs: internship level 1(1,2).ipynb - Co, Ask Gemini

URL: colab.research.google.com/drive/1hW8UZh9mGZeKsu4nUWAv3bPAT21tArK#scrollTo=y3TBMalqIDDj

Code cell [31] 0s:

```
plt.plot(
    [minimum, maximum],
    [minimum, maximum],
    linewidth=2
)

plt.title("Actual vs Predicted Journey Duration")
plt.xlabel("Actual Journey Duration (Minutes)")
plt.ylabel("Predicted Journey Duration (Minutes)")
plt.grid(True)

plt.tight_layout()

plt.savefig("Actual_vs_Predicted_Journey_Duration.png")

plt.show()

sample = pd.DataFrame({
    "Total_Distance": [350],
    "Number_of_Stops": [8]
})

predicted_duration = model.predict(sample)

print("\n===== SAMPLE JOURNEY PREDICTION =====\n")
print("Total Distance :", sample.loc[0, "Total_Distance"], "km")
print("Number of Stops :", sample.loc[0, "Number_of_Stops"])
print("Predicted Journey Duration :", round(predicted_duration[0], 2), "Minutes")

print("\nFinal prediction saved as 'Final Predictions.csv'")
print("Graph saved as 'Actual_vs_Predicted_Journey_Duration.png'")
print("\nTask 6.1 Completed Successfully.")
```

Bottom bar: Variables, Terminal, 12:12 AM, Python 3





